Dynamic Planet C - Dynamic Planet - Yale University Invitational 2022 - 02-12-2022

1. (3.00 pts) For the following six images, please provide the name for the lake type. End each entry with "lake," i.e. "man-made lake."

Word bank: oxbow lake, glacial lake, solution lake, volcanic lake, tectonic lake, landslide lake

1. 

2. 

3.



Volcanic lake | Glacial lake | Oxbow lake

2. (3.00 pts)

4.



5. 

6. 

Tectonic lake | Landslide lake | Solution lake

For Questions 7-16, please match the following drainage patterns with the corresponding catchment area description. **Please ensure your spelling exactly matches the list below or our software will mark the question as incorrect**.

Word bank: Trellised, Centripetal, Parallel, Herringbone, Dendritic, Rectangular, Angular, Radial, Barbed, Pinnate

3. (1.00 pts) 7. The rock type must be non-porous and impervious

Dendritic

4. (1.00 pts) 8. This drainage pattern is characteristic of folded mountains, like the Appalachian Mountains

Trellised

5. (1.00 pts) 9. This pattern originates from a hill with flow in all directions; Amarkantak is a notable example

Radial

6. (1.00 pts) 10. Drainage follows a regional slope and is more frequent in newly emerged coastal plains

Parallel

7. (1.00 pts) 11. This drainage pattern forms when streams converge at a point, typically a basin

Centripetal

8. (1.00 pts) 12. This pattern is formed when rocks are eroded along joints or fractures, resulting in clefts or rills used in draining

Rectangular

9. (1.00 pts) 13. This pattern usually develops in mountain-containing regions with steep slopes and valleys, where tributaries and master streams converge

Herringbone

10. (1.00 pts) 14. This drainage resembles the pattern of a leaf

Pinnate

11. (1.00 pts) 15. This drainage pattern originated when tributaries flowed in an opposite direction to the master stream

Barbed

12. (1.00 pts) 16. This drainage pattern is known as the “circular pattern”

Angular

Multiple Choice.

Please select the best answer from the given options.

13. (1.00 pts) 17. What is the largest freshwater lake by volume?

- A) Lake Superior
- B) Lake Baikal
- C) Lake Tanganyika
- D) Lake Victoria

14. (1.00 pts) 18. Which type of fan displays sheetflooding?

A) Eluvial
B) Colluvial
C) Fluvial
D) Alluvial

15. (1.00 pts) 19. Which term applies to the effect of wells on adjacent stream flow?

A) Well interference
B) Induced recharge
C) Gradient flow
D) Geohydrologic interference

16. (1.00 pts) 20. Which form of wetland has sedges as its dominant vegetation?

A) Marshes
B) Bogs
C) Fens
D) Swamps

17. (1.00 pts) 21. What is responsible for the less acidic pH of fens?

A) Frequent flowthrough of water
B) Accumulation of freshwater in the form of rain
C) Release of hydroxyl anions from alkali sediment
D) Presence of microorganisms capable of secreting natural buffer

18. (1.00 pts) 22. Oxbow lakes can be found in valleys with which type of central river?

A) Young
B) Adolescent
C) Mature
D) Old

19. (1.00 pts) 23. Which acid is primarily responsible for karst formation?

A) Hydrochloric acid (HCl)
B) Sulfuric acid (H_2SO_4)
C) Nitric acid (HNO_3)
D) Carbonic acid (H_2CO_3)

20. (1.00 pts) 24. Which of the following is not an example of an ecological benefit of tidal marshes?

A) Sheltering migratory birds
B) Sustaining a vast array of plant life
C) Ensuring proper nutrient regulation in outflowing water
D) Serve as a sanctuary for juvenile fish, clams, and crabs

21. (1.00 pts) 25. Pocosins are examples of:

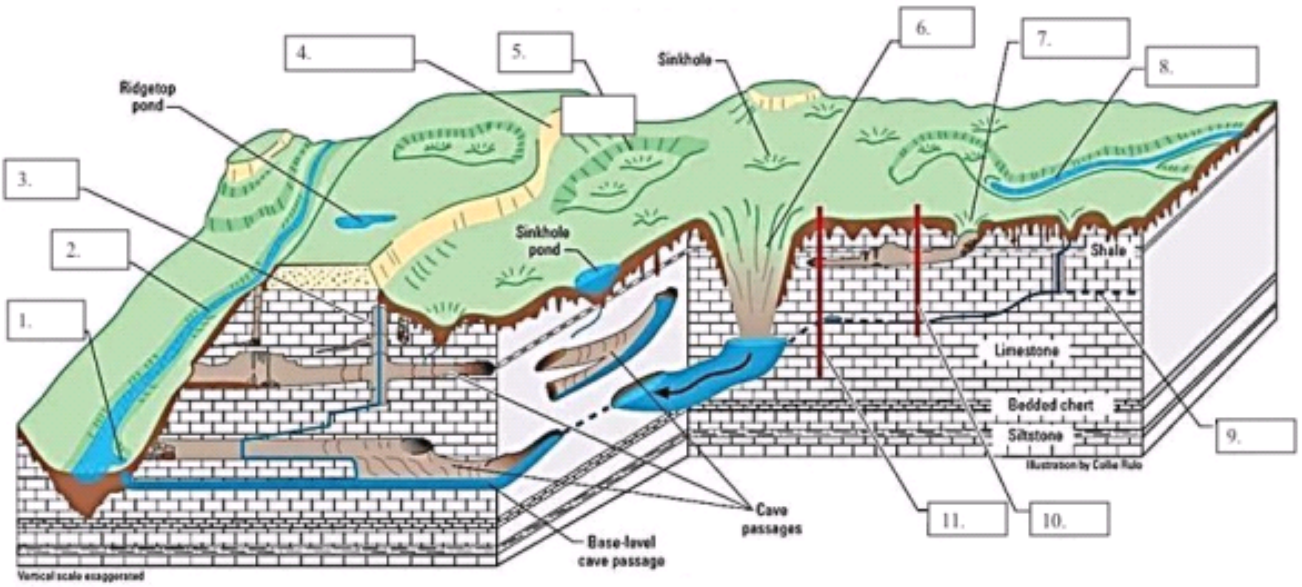
A) Rheotropic environments
B) Minerotropic environments
C) Ombrotropic environments
D) Phreatotropic environments

22. (1.00 pts)
26. A mountain is formed with alternating bands of weak and resistant bedrock. If a stream was found on this mountain, what drainage pattern would it most likely have?

A) Conical
B) Radial
C) Dendritic
D) Trellis

23. (1.00 pts)

27.



What process is displayed in the diagram above?

A) Cave formation
B) Groundwater entrapment
C) Caprock formation
D) Karst formation

Please label the above diagram. Some words may not be used. **Please ensure your spelling exactly matches the list below or our software will mark the question as incorrect.**

Word bank: Protective caprock, water table, stalagmites, Karst window, vertical shaft, sinking stream, cave entrance, base-level spring, dry hole, Karst valley, base-level stream, stalactites, producing well

24. (3.00 pts) *Please use the fill-in-the-blank lines below to indicate your responses.*

1.

2.

3.

Base-level spring | Base-level stream | Vertical shaft

25. (3.00 pts) 4.

5.

6.

Protective caprock | Karst valley | Karst window

26. (3.00 pts) 7.

8.

9.

Cave entrance | Sinking stream | Water table

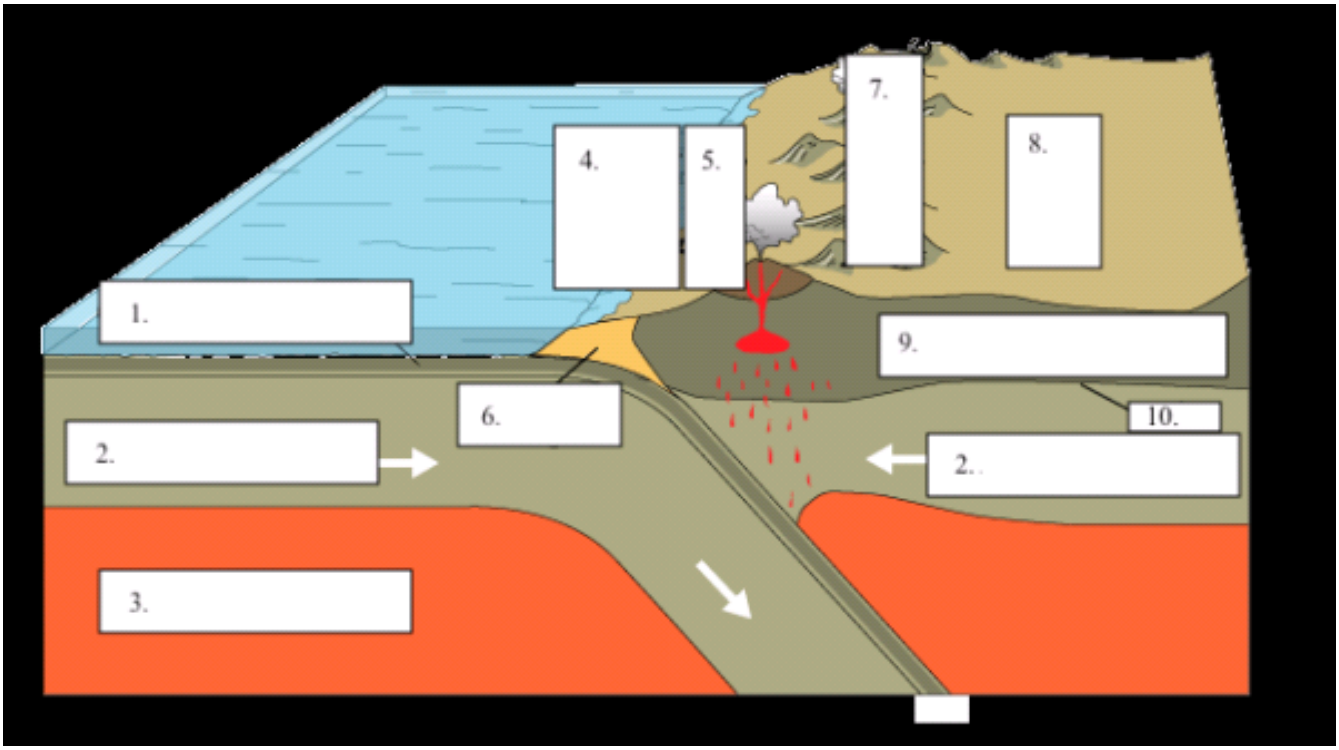
27. (2.00 pts) 10.

11.

Producing well | Dry hole

28. (1.00 pts)

28.



What process is displayed in the diagram above?

- (●) A) Subduction
- ( ) B) Orogenic belts
- ( ) C) Obduction
- ( ) D) Divergence

Please label the above diagram. Some words may not be used. **Please ensure your spelling exactly matches the list below, or our software will mark the question as incorrect.**

Word bank: Accretionary wedge, asthenosphere, lithosphere, forearc, moho, backarc, trench, oceanic crust, continental crust, volcanic arc, oceanic crust, mantle

29. (3.00 pts) *Please use the fill-in-the-blank lines below to indicate your responses.*

1.

2.

3.

Oceanic crust	Lithosphere	Asthenosphere

30. (3.00 pts) 4.

5.

6.

Trench	Forearc	Accretionary wedge

31. (3.00 pts) 7.

8.

9.

Volcanic arc | Backarc | Continental crust

32. (1.00 pts) 10.

Moho

33. (1.00 pts) 29. Which reservoir has the greatest residence time?

- A) Atmosphere
- B) Lithosphere
- C) Moho
- D) Biosphere

34. (1.00 pts) 30. What type of rock stands at the intersection of the *two* circuits of the Rock Cycle?

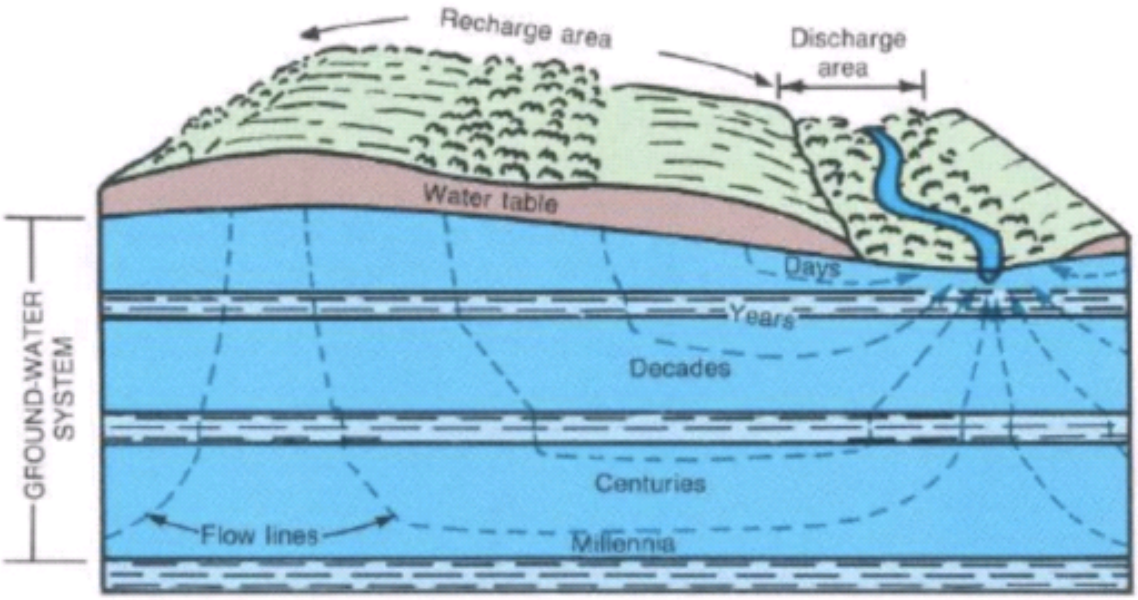
- A) Metamorphic
- B) Sedimentary
- C) Igneous
- D) Primitive

35. (1.00 pts) 31. Which of the following is not a function of a piezometric surface?

- A) Indicating direction of groundwater flow
- B) Determine hydraulic gradients
- C) Indicate the head of an aquifer
- D) Reveal the maximum datum plane

36. (1.00 pts)

32.



What type of aquifer is represented by the diagram above?

- (●) A) Confined
- ( ) B) Unconfined
- ( ) C) Perched
- ( ) D) Leaky

Focusing on Freshwater:

37. (1.00 pts) 33. What percentage of Earth's freshwater is available for animal and human use?

Expected Answer: 0.008%

38. (1.00 pts) 34. Freshwater is defined as having a concentration of less than how many ppt of dissolved salts? *Ocean water contains ~35,000 ppm of dissolved salts.*

Expected Answer: Less than 1-2 ppt (do not accept ppm)

39. (6.00 pts) 35. (1) Please list the three most abundant sources of freshwater on Earth.

(2) Provide the relative percentages of Earth' total water supply that each source of freshwater supplies.

Expected Answer: In glaciers: 1.717% In ground: 0.750% Frozen: 0.025%

Studying Steams:

40. (6.00 pts)
36. Jeff takes a leisurely ride down a river on his brand new raft and records a total distance of 768 meters as he floats down. From Jeff's starting point, a bird flies in a straight line fo 243 meters and ends up at the same place as Jeff.

(1) What is the sinuosity index of the stream? Please round your answer to the nearest thousandths.

(2) How would you describe the stream?

Expected Answer: Sinuosity of 3.16 (768/243) Braided (also accept: anastomosing)

41. (6.00 pts) 38. Manning's roughness coefficient is dependent on the following (select all that apply):

(Mark **ALL** correct answers)

- [x] A) Surface roughness
- [ ] B) Depth
- [x] C) Vegetation
- [ ] D) Stream capacity
- [ ] E) Dissolved ions
- [x] F) Cross-section irregularities

42. (3.00 pts) 37. Jeff travels to the Nile river to test his brand new raft. According to the sinuosity index, how would you describe the river?

Expected Answer: Meandering

Groundwater & Aquifers:

43. (1.00 pts) 39. Which of the following is *not* a volatile organic compound (VOC)?

- ( ) A) Chloroform
- ( ) B) Perchloroethylene (PCE)
- (•) C) Pentachlorophenol

D) Acetone

44. (1.00 pts) 40. Which of the following remediation techniques does not address VOC accumulation?

A) Permeable reactive barriers (PRB)
B) Air sparging
C) Air stripping
D) Chemical oxidation and reduction

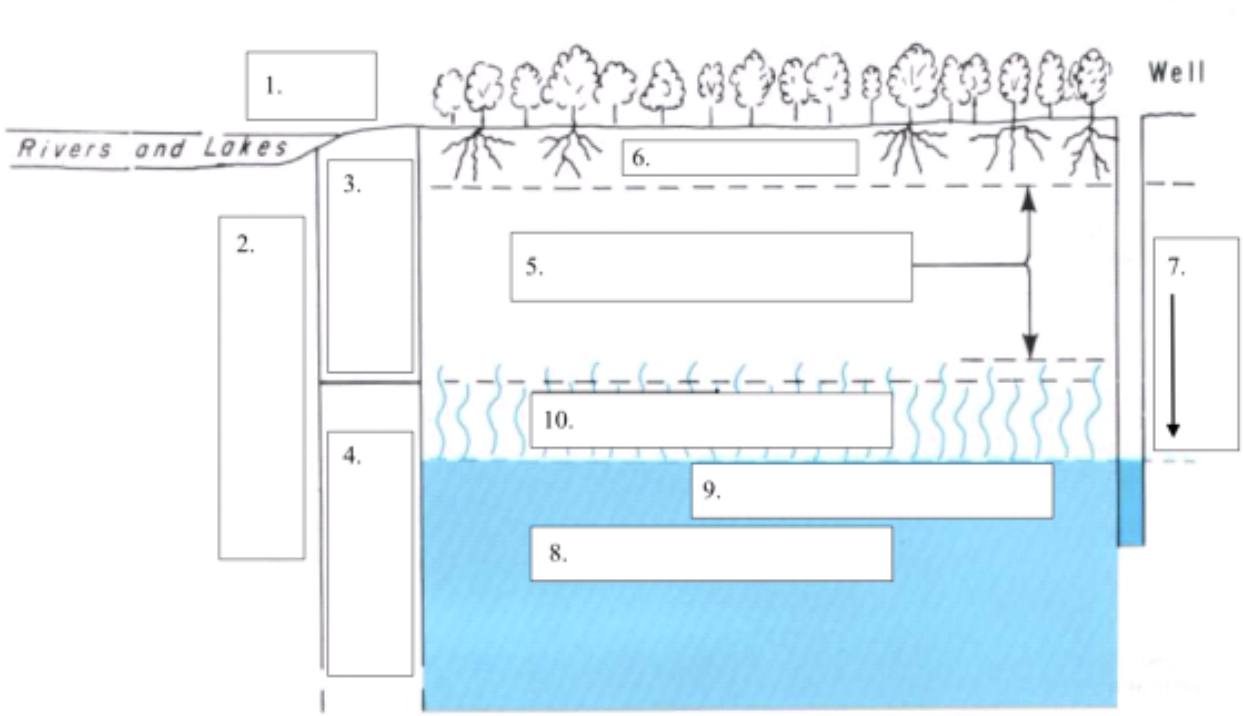
45. (1.00 pts) 41. The largest aquifer in the United States is the Ogallala Aquifer, in what aquifer system is it part of?

A) The High Plains
B) Central Valley
C) Columbian Plateau
D) Great Basin Carbonate

46. (1.00 pts) 42. What is the primary factor for determining the depth of a cone of depression?

A) Proximity to bodies of water
B) Pumping rate
C) Well depth
D) Soil porosity

43.



Please label the above diagram. Some words may not be used. **Please ensure your spelling exactly matches the list below or our software will mark the question as incorrect.**

Word bank: Unsaturated zone, ground water, saturated zone, surface water, water level, water table, peat zone, intermediate zone, capillary fringe, soil zone, underground water, semi-permeable zone

47. (3.00 pts) 1.

2.

3.

Surface water	Underground water	Unsaturated zone

48. (3.00 pts) 4.

5.

6.

Saturated zone	Intermediate zone	Soil zone

49. (3.00 pts) 7.

8.

9.

Water level	Ground water	Water table

50. (1.00 pts) 10.

Capillary fringe

51. (2.00 pts) 44. (1) Please distinguish between permeability and hydraulic conductivity.

(2) Provide the formula for conversion from hydraulic conductivity and permeability.

Expected Answer: (1) Permeability = intrinsic property of pore structure Hydraulic conductivity = depends on the properties of the fluid (viscosity, temperature, density, saturation, etc) (2) Multiply k by fluid viscosity. Divide this value by fluid density and the gravitational constant.

52. (3.00 pts) 45. Please describe three different approaches taken to estimate the hydraulic conductivity.

Expected Answer: *Any three of the following works! Partial credit can be awarded. Laboratory tests: testing on core samples Visual test: visualizing the soil type and grading based on published values and/or prior experience Borehole tests: in-situ test, where boreholes are drilled into region and observed Particle size correlations: use empirical correlations to relate particle size in granular soil to a hydraulic conductivity value Pumping tests: controlled and monitored pumping from at least one well, followed by a record of the drawdown observed in wells of a given pumped flow rate

53. (6.00 pts) 46. Please describe the Primary (Natural), Secondary, and Tertiary types of groundwater salinity.

Expected Answer: Natural/primary salinity: contact with bedrocks containing highly soluble minerals or an accumulation of salts from rainwater over time Secondary/dry land salinity rising water levels bring salt to clearing of vegetation or salt to the surface, causing accumulation of salt in the soil Tertiary/irrigated salinity: increase in salt after multiple irrigations of water, wherein salt remains after evaporation, accumulates temporally, and leeches into the ground (2 points per Primary (Natural), Secondary, and Tertiary, all of nothing)

54. (3.00 pts) 47. Please describe the process by which surface water becomes groundwater.

Expected Answer: *Precipitation allows rainwater to fall from the atmosphere to the ground. *Infiltration occurs, allowing water to enter the ground. *Percolation occurs, whereby the water enters the water table. Look for the presence of the words with the * indicator. 1 pt each.

55. (3.00 pts) 48. (1) Please describe the significance of a confining layer.

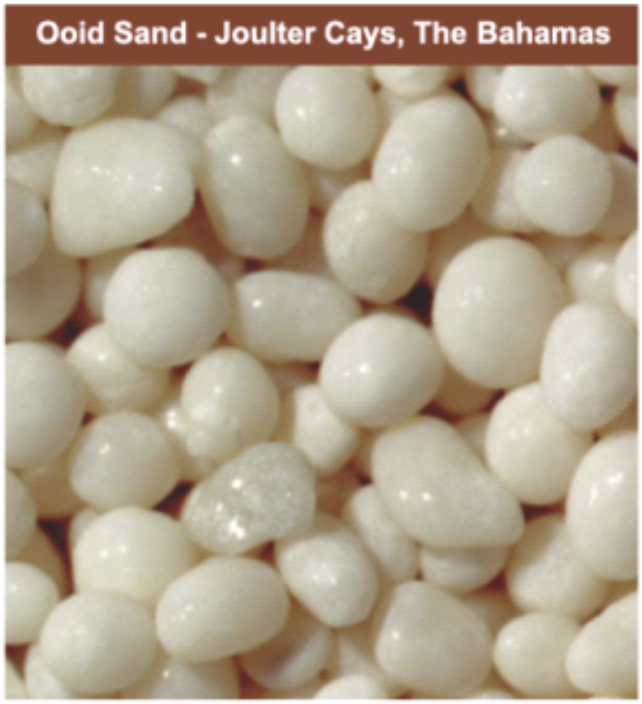
(2) Provide two geological materials that can make up a confining layer.

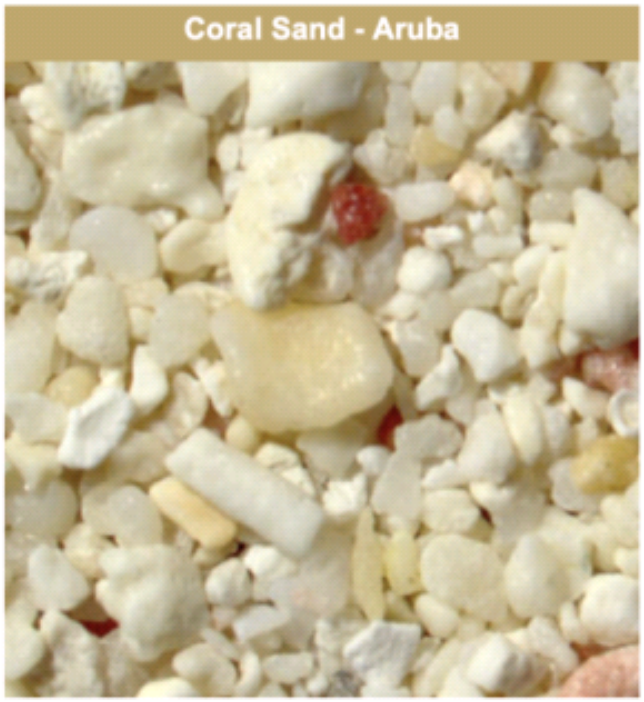
Expected Answer: A confining layer is a layer adjacent to an aquifer that does not allow for the flow of water. Two examples include any geological material that is “not consolidated, such as rock without holes, clay, mudstone, and siltstone.

56. (1.00 pts) 49. What is the significance of the Beyer formula?

Expected Answer: Estimates the hydraulic conductivity (k) from sediments’ grain distribution

57. (2.00 pts)





50. (1) Which sand layer will have the greatest void fraction?

(2) How do you calculate the porosity for a rock sample?

Expected Answer: Ooid Sand (1 pt) The porosity of a rock sample can be calculated by dividing the pore volume by the total volume (1 pt)

58. (3.00 pts) 51. (1) How does porosity vary from permeability?

(2) Can a substance be both porous and non-permeable? Explain your answer.

Expected Answer: Porosity = the measure of the “void” of a substance Permeability = a measure of a substance’s ability to transmit fluids (1 point for comparing permeability and porosity) (2 pt- one for Yes, one for proper description) Yes, if pores are not connected, a rock can be porous but not permeable.

General Questions:

59. (3.00 pts) 52. (1) Compare the oceanic crust with the continental crust.

(2) In which would one find granite? Basalt?

Expected Answer: The oceanic crust is much thinner than the continental crust (Basalt) The continental crust is slightly less dense than the oceanic crust. In addition, the continenta crust is much thicker and makes up ~ 70% of Earth’s crust volume (Granite) (1 pt for comparison, 1 pt for basalt - oceanic crust, 1 pt for granite - continental crust)

60. (3.00 pts) 53. Please list the three main layers in lake thermal stratification.

Please list the layers in order, from most surface-level to most deep.

Expected Answer: 1. Epilimnion 2. Metalimnion 3. Hypolimnion *Please ensure the answers are provided in that order.

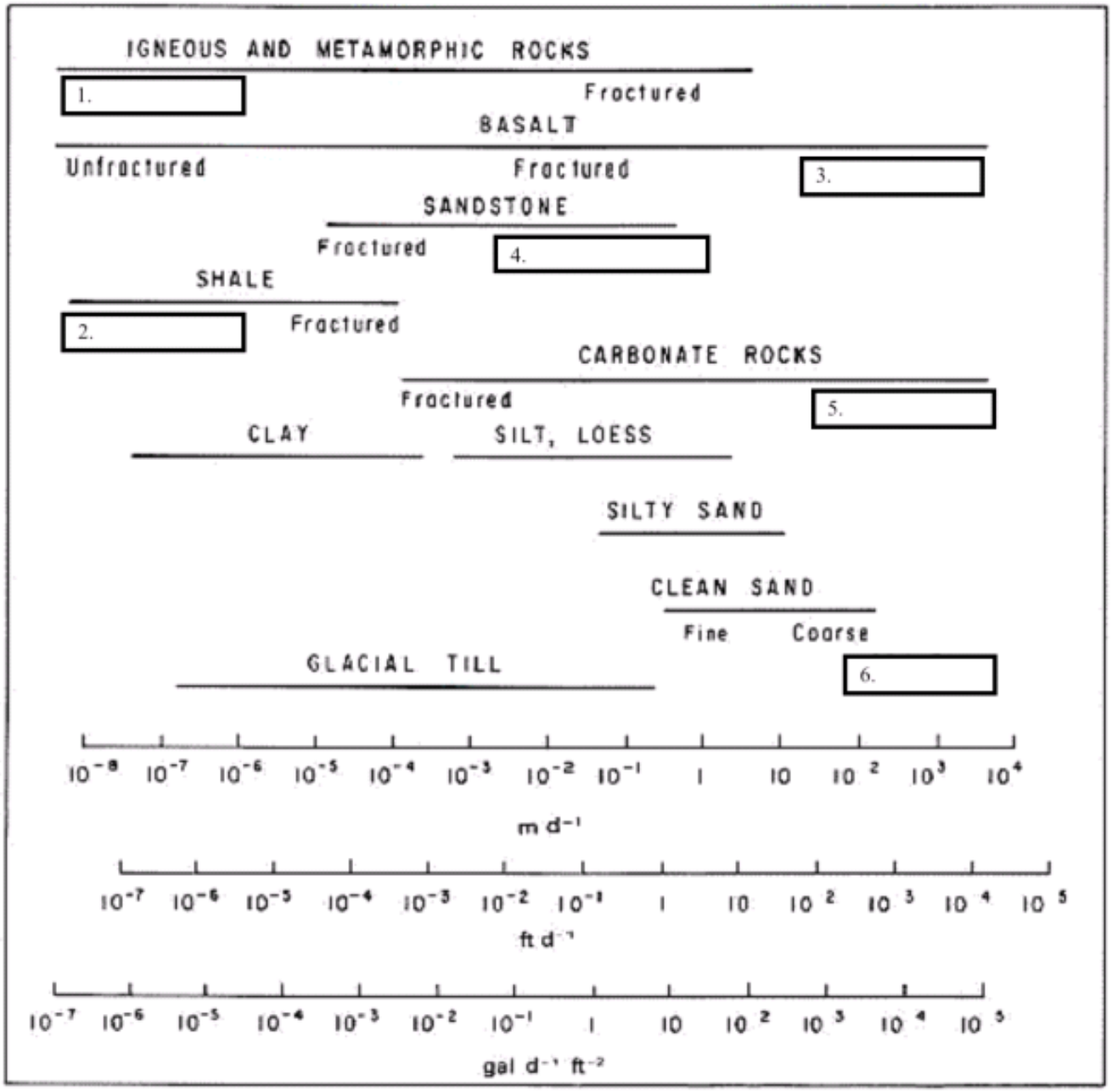
61. (3.00 pts) 54. Please provide the three major ways oxygen can enter a lake.

Expected Answer: Atmospheric diffusion Aquatic plants photosynthesize and release oxygen into the water Rivers and streams bring oxygenated water into the lake

62. (1.00 pts) 55. Please define a moraine.

Expected Answer: Material (typically soil and rock) left behind by a moving glacier.

Tie-Breaker



Please complete the above diagram displaying the hydraulic conductivity of rocks.

A word bank has not been given for this question.

63. (6.00 pts)

1.

2.

3.

4.

5.

6.

Expected Answer: 1) Unfractured 2) Unfractured 3) Lava flow 4) Unfractured 5) Cavernous 6) Gravel

Thank you for taking the Division C Dynamic Planet test!

We wish you luck with all your future Science Olympiad endeavors!

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